Metadata of the chapter that will be visualized in SpringerLink

Book Title	Advancements in Automation, Robotics and Sensing	
Series Title		
Chapter Title	A Soft, Multi-finger Pneumatic Gripper for Versatile and Safe Object Handling	
Copyright Year	2027	
Copyright HolderName	ICST Institute for Computer Sciences, Social Informatics and Telecommunications Engineering	
Corresponding Author	Family Name	**Sarvesh**
	Particle	
	Given Name	**R.**
	Prefix	
	Suffix	
	Role	
	Division	Department of Robotics and Automation Engineering
	Organization	PSG College of Technology
	Address	Coimbatore, 641004, Tamil Nadu, India
	Email	sarvesh.r.clg@gmail.com
	ORCID	https://orcid.org/0009-0000-4367-1953
Author	Family Name	**Parthiban**
	Particle	
	Given Name	**M.**
	Prefix	
	Suffix	
	Role	
	Division	Department of Robotics and Automation Engineering
	Organization	PSG College of Technology
	Address	Coimbatore, 641004, Tamil Nadu, India
	Email	parthiban272005@gmail.com
	ORCID	https://orcid.org/0009-0003-9127-4938
Author	Family Name	**Vijitha**
	Particle	
	Given Name	**K.**
	Prefix	
	Suffix	
	Role	
	Division	Department of Robotics and Automation Engineering
	Organization	PSG College of Technology
	Address	Coimbatore, 641004, Tamil Nadu, India
	Email	kvijitha942006@gmail.com
	ORCID	https://orcid.org/0009-0005-7389-2470
Author	Family Name	**Udhayamathi**
	Particle	

	Given Name	**S.**
	Prefix	
	Suffix	
	Role	
	Division	Department of Robotics and Automation Engineering
	Organization	PSG College of Technology
	Address	Coimbatore, 641004, Tamil Nadu, India
	Email	udhayamathi0706@gmail.com
	ORCID	https://orcid.org/0009-0000-6932-1943
Author	Family Name	**Vishwa Priya**
	Particle	
	Given Name	**V.**
	Prefix	
	Suffix	
	Role	
	Division	Department of Robotics and Automation Engineering
	Organization	PSG College of Technology
	Address	Coimbatore, 641004, Tamil Nadu, India
	Email	spk.rae@psgtech.ac.in
		vvp.rae@psgtech.ac.in
	ORCID	https://orcid.org/0000-0003-2493-7511

Abstract	This article introduces a soft 3-claw pneumatic gripper designed for adaptive and dexterous object manipulation. Liquid silicone is an elastic material, which is vital in handling delicate and non-uniform objects are incorporated in this gripper. This system can be utilized in a wide range of fields ranging from soft robotics in unstructured environments, industrial automation. Influenced by bionics, the design mirrors the principles of nature to increase functionality and efficiency, specifically taking cues from the tentacles of an octopus and plant tendrils, both reputed for their soft-bodied, flexible, and adaptive grasping properties. The octopus tentacle especially can wrap around objects of diverse shapes and sizes with great precision and evenly distributed force, a mechanism that is copied in this gripper in the form of pneumatic actuation and deformation of soft material. Bionics enhances the effectiveness of robotics and provides new avenues in human-interactive robotics with capabilities to design safer robots. With the use of biological morphology in soft robotics, this methodology allows for effective functioning in unstructured and sensitive environments and is a major advancement toward integrating robotics into more natural and variable applications.

Keywords (separated by '-')	Soft pneumatic gripper - flexible actuator - finite element analysis - 3D printing

Author Proof

Check for updates

A Soft, Multi-finger Pneumatic Gripper for Versatile and Safe Object Handling

R. Sarvesh(✉), M. Parthiban, K. Vijitha, S. Udhayamathi, and V. Vishwa Priya

Department of Robotics and Automation Engineering, PSG College of Technology, Coimbatore 641004, Tamil Nadu, India
sarvesh.r.clg@gmail.com, {spk.rae,vvp.rae}@psgtech.ac.in

Abstract. This article introduces a soft 3-claw pneumatic gripper designed for adaptive and dexterous object manipulation. Liquid silicone is an elastic material, which is vital in handling delicate and non-uniform objects are incorporated in this gripper. This system can be utilized in a wide range of fields ranging from soft robotics in unstructured environments, industrial automation. Influenced by bionics, the design mirrors the principles of nature to increase functionality and efficiency, specifically taking cues from the tentacles of an octopus and plant tendrils, both reputed for their soft-bodied, flexible, and adaptive grasping properties. The octopus tentacle especially can wrap around objects of diverse shapes and sizes with great precision and evenly distributed force, a mechanism that is copied in this gripper in the form of pneumatic actuation and deformation of soft material. Bionics enhances the effectiveness of robotics and provides new avenues in human-interactive robotics with capabilities to design safer robots. With the use of biological morphology in soft robotics, this methodology allows for effective functioning in unstructured and sensitive environments and is a major advancement toward integrating robotics into more natural and variable applications.

Keywords: Soft pneumatic gripper · flexible actuator · finite element analysis · 3D printing

1 Introduction

Soft robotics has emerged as a transformative area in engineering by enabling safe, adaptive, and effective interaction with fragile, smooth, irregular, and complex-shaped objects. Unlike conventional rigid robotic systems, the soft robotic grippers make use of highly deformable materials and flexible architectures inspired by biological organisms like octopus tentacles, which naturally possess far greater adaptability and conformal contact [6, 7]. These bio-inspired mechanisms have encouraged the use of elastic materials and complex structures in robotic manipulation, advancing the future toward greater safety and versatility in unstructured environments [1, 3].

© ICST Institute for Computer Sciences, Social Informatics and Telecommunications Engineering 2027
Published by Springer Nature Switzerland AG 2026. All Rights Reserved
Vinod B et al. (Eds.): ICAARS 2025, LNICST 719, pp. 1–16, 2027.
https://doi.org/10.1007/978-3-032-41656-8_5

Author Proof

Among different actuation approaches, pneumatic actuation has gained high popularity owing to its simplicity, low cost, ease of control, and allows for large, smooth deformations in soft materials [3, 9, 12, 14].

Although different chamber geometries and materials, as well as various actuator configurations, have been explored in multiple studies, challenges persist in the pursuit of an optimal balance between flexibility, load-bearing capacity, fabrication accuracy, and repeatability. Many existing soft pneumatic grippers show high compliance with limited structural support for moderate-weight objects [3, 8]. Fabrication-related issues include air gaps, uneven chamber deformation, and surface defects, which commonly occur in single-step molding or direct multi-material 3D printing approaches [1, 2]. Additionally, limited detailed simulation integration restricts predictive insight into deformation patterns, strain distribution, and failure modes, and hence suboptimal performance of actuators is realized [6, 9, 13]. Overall, these gaps indicate a need for grippers that are not only adaptable but also structurally reliable and accurately modeled.

In this research, emphasis is placed on the design and fabrication of a soft pneumatic three-claw gripper capable of grasping objects with diverse shapes and sizes. Highly elastic and resilient material, liquid silicone, is used to improve flexibility and durability [4, 12]. A hybrid fabrication method combining 3D printing and injection molding is adopted to ensure precise chamber geometry, airtight construction, and high repeatability, addressing limitations reported in earlier techniques [1, 2]. Finite element analysis (FEA) is used to predict deformation behavior, strain profiles, and kinematic response under varying pressure inputs, enabling optimization of chamber design and material distribution prior to prototyping [6, 9, 13]. The proposed system contributes to current soft robotics research by presenting a bio-inspired, structurally reliable, and customizable gripper design validated through fabrication and experiment.

This work represents a hybrid two-stage molding technique in fabricating multi-layered soft fingers with embedded chambers. The method achieves higher dimensional accuracy, improved airtight sealing, and repeatability better than 2% by separating internal and external mold formation, while outperforming the conventional one-shot casting and low-cost 3D-printed actuators. A bio-inspired geometry, based on octopus tentacle actuation, results in a three-claw with increased structural rigidity-a rather uncommon combination within the spectrum of soft pneumatic grippers. This design allows for the reliable handling of both delicate and heavy objects.

It also introduces a cross-validated FEA approach that are rarely demonstrated for low-cost soft robotic systems, and they allow a more reliable understanding of actuator performance. Material novelty is provided here by a hybrid TPU–silicone architecture; the intrinsic actuation material, LSR-105 silicone, provides flexibility, while the embedded TPU components add stiffness and print fidelity. This provides for fully fatigue-free actuation over > 200 cycles, enabling durability and control not achievable from simple-material silicone actuators.

2 Literature Review

Recent advancements in soft robotics have led to the development of flexible grippers that mimic biological structures for safe and adaptive object manipulation. Researchers have explored various materials, geometries, and actuation methods, particularly pneumatic

Author Proof

systems, to enhance grasping efficiency. Studies emphasize the importance of simulation tools such as finite element analysis for performance prediction and structural optimization. This section reviews key contributions in soft gripper design, focusing on material innovation, actuation strategies, and bio-inspired configurations.

Hybrid fabrication techniques combining 3D printing and molding processes have been shown to enable the development of customized soft actuators with improved design flexibility [1]. The mechanical design and experimental evaluation of flexible pneumatic grippers highlight the importance of material flexibility in achieving high conformability during grasping [2]. Recent approaches incorporating sensor-based feedback control systems have demonstrated improvements in gripping accuracy and motion reproducibility, particularly through PWM-based airflow control [3]. Material advancements involving thermoplastic polyurethane (TPU) and silicone have further enhanced elasticity and durability in soft gripper designs [4].

Bio-inspired design trends have gained significant attention, with soft grippers increasingly modeled after natural systems such as the octopus and human hand [5]. Studies on actuator kinematics and deformation modeling have emphasized the role of simulation tools, such as Abaqus, in achieving predictable and optimized structural responses under pneumatic loading conditions [6]. Comprehensive reviews of soft gripper technologies highlight ongoing developments in material innovation, actuation mechanisms, and bio-inspired configurations for improved human–robot interaction [7]. Structural optimization and experimental validation approaches have demonstrated the importance of geometry and pressure-based deformation in enhancing grasping performance [8].

The design and performance of soft pneumatic actuators have also been widely studied, particularly emphasizing the use of silicone-based materials for controlled actuation and high conformability [9]. Low-cost hardware-based control strategies have enabled the deployment of pneumatic grippers in affordable robotic systems for unstructured environments [10]. Recent work on optimized soft pneumatic grippers with integrated control systems has shown improved response time and grip accuracy using PWM-based actuation techniques [11]. These studies collectively indicate a trend toward adaptive, integrated, and simulation-driven soft gripper designs. The present work aims to address existing gaps by integrating simulation, fabrication, and experimental validation into a unified optimized soft gripper system.

3 Methodology

The creation of the soft 3-claw pneumatic gripper was executed based on a systematic process that involves design, material choice, hybrid manufacturing, and experimental verification. The approach was based on the requirement for an efficient and flexible gripping device able to handle fragile and inhomogeneous objects by means of pneumatic actuation. The first phase was the conceptual and parametric design of the gripper using computer-aided design (CAD) software Autodesk Fusion 360. The gripper has three flexible fingers symmetrically disposed over a central base, each of which has internal pneumatic chambers. These enable controlled bending on pressurization, allowing for a conforming grip on objects. The design helped to optimize geometry to obtain maximum bending displacement with low material strain which guarantees both performance and

Author Proof

durability. Material choice was important to obtain the required performance. Liquid silicone rubber (LSR) was chosen for the finger's fabrication because of its high elasticity, low shore hardness, and biocompatibility. It can be used in fields of soft actuation because of its hyper-elastic nature.

A hybrid manufacturing method can be utilized to integrate the strengths of additive manufacturing as well as conventional molding. The negative moulds of the fingers could print using 3D printing technology with polylactic acid (PLA) material. Next, the printed molds can be employed in a manual injection molding process, where deaerated liquid silicone was poured and cured to shape the soft actuators. This method provides high repeatability, dimensional accuracy, and the capacity to replicate intricate internal details like embedded air channels. The hybrid manufacturing process guarantees finger modules of high quality with few surface defects. The PLA molds 3D printed enables precise control of the internal channel geometry. The silicone injection ensures uniformity in dimensions and airtight sealing. Actuators show high repeatability, having less than 2% deviation in dimensions from unit to unit, could prove the validity of the hybrid method for batch fabrication. Finite element simulations were employed for predicting and optimizing the deformation response of the gripper. The silicone hyper-elastic material model was calibrated and implemented to model the response of the actuator to different pneumatic pressures. The simulations tested total displacement, total strain in each axis and are displayed below. Overall, this approach combines mechanical design, high-performance materials, manufacturing and computational simulation, into a unified platform for soft robotics R&D. The result is a complete working soft pneumatic gripper capable of executing applications from industry to hospital.

4 Design and Development of Gripper

The development and design of the soft pneumatic gripper were achieved through a systematically designed sequence of stages such as digital modeling, material processing, molding, and assembly. The aim was to develop a low-cost, flexible, and adaptable gripper that could be used to manipulate objects of different shapes, sizes, and fragility nature. The overall strategy involved the emphasis on simplicity, and compatibility with readily available fabrication techniques such as 3D printing.

4.1 CAD Design and Customization

The process started with the downloading of readily available 3D models. The available files contained necessary parts such as inner cavity molds for mounting the soft fingers, outer molds to shape the silicone cover, and a base plate for mounting fingers. The downloaded files were then adapted using Autodesk Fusion 360 to fit the intended use. Changes consisted of altering finger length, curvature, and wall thickness to enhance flexibility and performance with pneumatic actuation. Further integration points were incorporated to provide openings for pneumatic tubing and fit securely with the air connectors. Complementary 3D-printable components were also created to form internal threads (to facilitate air supply into the claws) and alignment features to help with post-molding assembly.

Author Proof

4.2 Mold Fabrication and Material Preparation

All parts were printed with PLA filament, chosen for its ease of printing and adequate mechanical strength in mold applications. The print resolution was maximized to provide smooth surfaces and sharp sealing edges where required. The base mold (Fig. 1) dictates the bottom curvature and structure of each claw including guide rails and bottom sealing features. Center mold (Fig. 2) contains minute internal details like the chamber patterns and rib structures responsible for pneumatic actuation. The upper mold (Fig. 3) closes off the cavity by covering up the top geometry of the claws and sealing the mold during silicone casting. In combination, these molds facilitate precise and reproducible fabrication of the soft robotic gripper from materials.

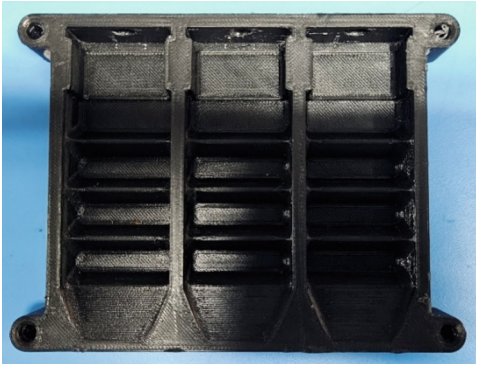


Fig. 1. Base mold showing lower cavity geometry used to form finger curvature

The manufacturing process utilized a two-stage molding approach with food or medical-grade silicone rubber to achieve biocompatibility as well as flexibility in gripper construction. The LSR105 is the cure liquid silicone rubber used for making skin soft material making. It cures to an elastic rubber of high strength and translucency at room temperature (Table 1). AQ4

LSR105 is a two-part compound, Part A and Part B, which were combined in equal volumes of 1:1 as shown in Fig. 4. The parts were mixed well for about 4 min to create a uniform mixture necessary for uniform curing and mechanical performance. To get rid of trapped air bubbles prior to casting, two methods of degassing were applied. Preferably, a vacuum chamber was employed to set up a vacuum condition for 3–5 min, allowing effective draining out of air pockets. Without a vacuum facility, the silicone composition was allowed to rest in room air for approximately 1 h to allow bubbles to rise and escape naturally. This precautionary planning assured the highest structural integrity and surface finish of the molded soft actuator parts.

4.3 Two-Stage Molding Process

The molding process was segmented into two major stages. Internal cavity mold was constructed, and the degassed silicone blend was poured slowly into it to avoid bubble

Author Proof



Fig. 2. Centre Mold for Internal Chamber Formation



Fig. 3. Top Mold for Outer Layer Closure

Table 1. LSR-105 Material Properties

Description	Parameter	Unit
Hardness	05	A
Density	1100	kg/m^3
Viscosity	3000 (At 25 °C)	cps
Tensile strength	≥35	kgf/cm^2
Modulus of Elasticity	0.10	MPa
Poisson's Ratio	0.495	–
Shear Modulus	0.033	MPa
Bulk Modulus	150	MPa

Author Proof

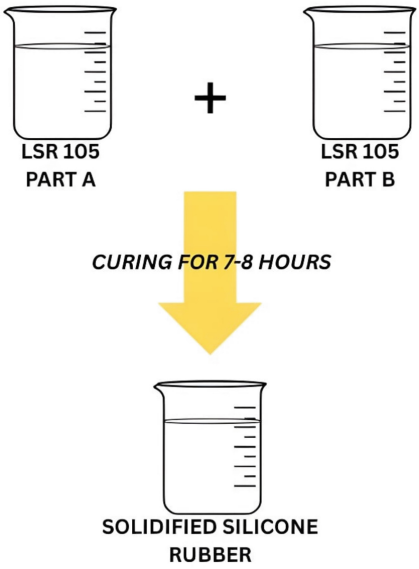


Fig. 4. Mixing and Curing of LSR-105 Silicone

entrapment. The mold was tightly closed using clamps or tape to provide the correct curing geometry. After room temperature curing for 6 to 8 h, the mold was opened, and the inner finger structure was left enclosed in the cavity as shown in Fig. 5.

At the second stage, further 3D-printed parts were bonded to the partially cured silicone to create air inlets and connectors. Inner finger, which was molded previously, was placed inside the outer mold. From Fig. 6, Another group of silicone was prepared and poured over the internal core to create the outer layer. This arrangement was sealed once again and allowed to be cured for another 6 to 8 h. The mold was opened gently after completion to show the final soft finger with internal air channels and flexible form. This two-layer strategy enabled tailored internal cavities with the assurance of strong external support for pneumatic deformation.

4.4 Final Assembly and System Integration

After curing and demolding, air connectors were placed in pre-formed inlet channels within every soft finger. Silicone adhesive was applied for joints to provide an airtight seal to prevent any leakage during use. The soft fingers were then attached to a 3D-printed base plate using screws, locking tabs, or adhesive, based on design. The base plate was also tailored to feature routing for pneumatic tubing. All the fingers were interfaced with a pneumatic air supply system through a Y-connector to provide even distribution of air. The system accommodated both positive and negative pressure inputs, obtained through a vacuum pump or an air compressor, through inflation and deflation cycles. A manual - controlled system was employed to control airflow for dynamic actuation as shown Fig. 8.

Author Proof



Fig. 5. Stage 1 Silicone Casting (Inner Structure)



Fig. 6. Stage 2 Silicone Casting (Outer Encapsulation)

The model, from Fig. 7 showed considerable grasping ability for different objects, from delicate objects such as eggs to flexible materials. The design also remained compatible with robotic arms and conveyor belts, making it adaptable for various real-world applications.

5 Results and Testing

A series of experimental tests were undertaken to test the performance of the prototyped soft 3-claw pneumatic gripper. All subsystems were separately and jointly tested to prove its behavior under different operating conditions.

5.1 Flexible Finger Actuation and Material Response

The silicone fingers were tested with incremental pneumatic pressures, in which each finger showed smooth and repeatable bending behavior. The response was non-linear and consistent with the hyper-elastic properties of the silicone material. The fingers returned

Author Proof



Fig. 7. Assembled 3-Claw Pneumatic Gripper

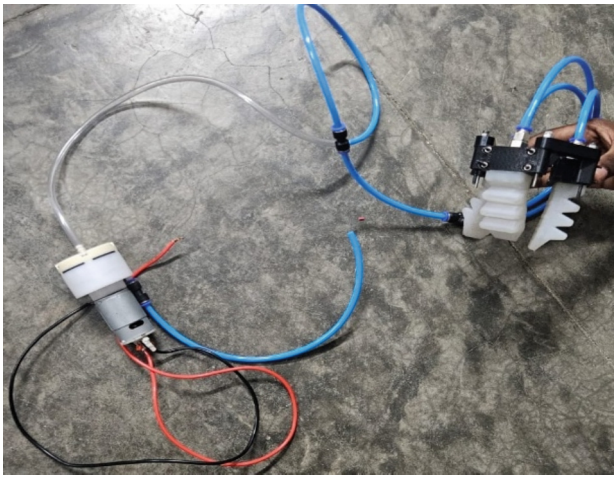


Fig. 8. Pneumatic circuit setup

to their original shape without residual deformation which is an indication of good elasticity and fatigue life over many cycles. The implementation of LSR made sure that there was consistency throughout the trials. There was also no observable degradation due to the repeated actuations up to a cycle of 200. Tests on the integration of the system ensured that the coordination between the actuators was smooth. The system that was developed was able to accommodate programmable grasping. Tests on the repeatability aspect ensured that the retaining force on the object was stable after a cycle of 100 actuations (Tables 2, 3, and 4).

Author Proof

5.2 Testing Soft Gripper Properties

Table 2. Structural material properties assigned to finger

Properties	Values	Unit
Density	1.10	g/cm^3
Modulus of Elasticity	0.10	MPa
Poisson's Ratio	0.495	–

Table 3. Pressure vs. Kinematics

Pressure (bar)	Bending angle	Tip displacement (mm)
0.10	12.5	4.7
0.15	18.6	7.1
0.20	24.8	9.5
0.25	30.5	12.1

Table 4. Pressure vs. Gripping Force

Pressure (bar)	Max held weight (g)	Gripping force (N)
0.10	43	0.39
0.15	69	0.64
0.20	86	0.80
0.25	110	1.03

5.3 FEA Methodology

The silicone actuator (LSR-105) was modeled for analysis as a near-incompressible hyperelastic material based on the third-order Yeoh model, expressed as:

$$W = C_1(I_1 - 3) + C_2(I_1 - 3)^2 + C_3(I_1 - 3)^3,$$

where I_1 denotes the first invariant of the right Cauchy-Green deformation tensor. Poisson's ratio of $\nu = 0.495$ was assigned to account for near incompressibility. These parameters were identified based on nonlinear regression of the uniaxial tensile and compression test data, for which probability for goodness of fit was assessed via R^2. The embedded TPU reinforcement zones were modeled as linear elastic materials using E = 30 MPa and $\nu = 0.33$, given that strain levels remained in the linear during actuation.

Author Proof

A detailed 3D model of the finger, including all multi-layer walls and internal pneumatic chambers, was used for simulation. Quadratic tetrahedral elements using a hybrid formulation were chosen to better represent large deformations and stress concentrations.

A nonuniform tetrahedral mesh is used to discretize the actuator geometry, providing a balance between accuracy and computational efficiency. A global element size of approximately 1.5 mm was applied throughout, whereas critical regions like internal chambers, fillets, and thin wall sections were locally refined to 0.6 mm to capture steep strain gradients. The final mesh contained approximately 120,000–180,000 elements. Mesh independence was verified through a convergence study using element sizes of 2.5 mm, 1.5 mm, and 0.8 mm, where the difference in predicted tip displacement between the two finest meshes remained below 1.5%, confirming that the chosen mesh density provided sufficient accuracy for nonlinear deformation analysis.

The simulation proceeded with a detailed 3-D finger model representing all multi-layer walls and internal pneumatic chambers. For capturing large deformation and providing stress concentration accurately, the quadratic tetrahedral elements with hybrid formulation have been employed.

The actuator was constrained at the proximal mounting face with all DOFs fixed to represent its rigid attachment to the gripper base. Tied constraints were assigned to bonded interfaces. Regions with potential sliding were treated as frictionless contact. Pneumatic actuation was modeled as a uniform internal pressure applied to the cavity surfaces with a quasi-static ramp from 0.10 bar to 0.25 bar with an increment of 0.05 bar, modifiable to meet experimental conditions.

5.4 Analysis of Deformation, Stress and Strain

Comparison of both gripper structures in terms of total deformation presents opposing behaviors of flexibility and stiffness. The rectangle-shaped gripper presents a maximum deformation value of 50.428 mm, showing extremely high compliance with high bending mostly localized at the free tip, making it very well suited for soft robotics applications where delicate and conformal grasping is needed. On the other hand, our design has a very low maximum deformation of 0.0028805 mm, deformation distributed smoothly with very little overall deflection, indicating its stiffness and better load-bearing nature (Figs. 9 and 10).

AQ5

Author Proof

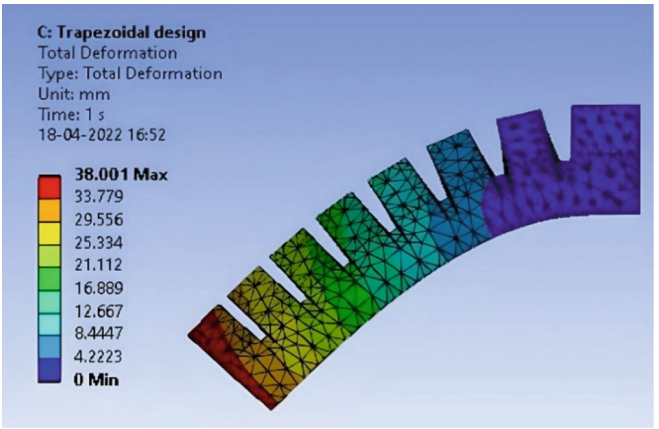


Fig. 9 (a) Deformation Comparison: Rectangular Type Designs.

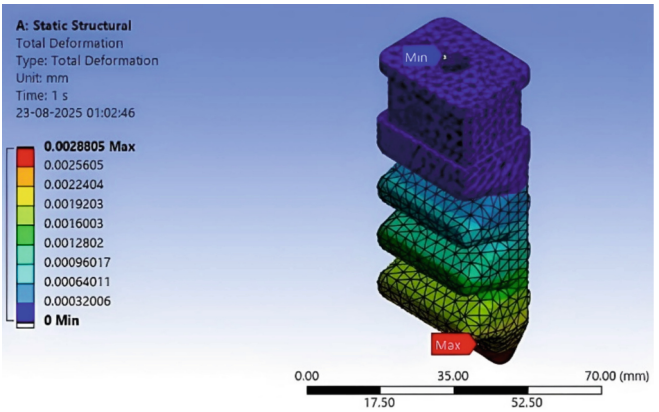


Fig. 10. (b) Deformation Comparison: Bellow-Type Designs.

Although the rectangle design is more applicable to fragile object handling owing to its flexibility, our design is more suited to applications requiring rigidity and stability to hold heavier loads (Table 5).

Table 5. Performance Comparison Between Current Design and Existing Design

Category	Existing observation	Current observation
Material Model	E = 50 MPa (too stiff for soft actuators).	E = 0.10 MPa, ν = 0.495 (for LSR-105).
Bending angle	50° at 60 kPa	63.8° at 50 kPa
Bending efficiency	0.83°/kPa	1.28°/kPa

(*continued*)

Author Proof

Table 5. (*continued*)

Category	Existing observation	Current observation
Tip displacement	50.4 mm (longer finger)	28.5 mm at 50 kPa (shorter finger)

Material properties for soft actuators are more realistic in the present observation compared to the previous work, where it applied a stiff silicone model characterized by a Young's modulus of 50 MPa, while this work applies the LSR-105 parameters: $E = 0.10$ MPa and $\nu = 0.495$, hence giving a more realistic modeling of ultra-soft pneumatic actuators and providing more reliable deformation predictions. In the light of such bending performance, the previous actuator realized about 50° at 60 kPa, while the actuator developed herein reached 63.8° at the lower pressure of 50 kPa. Correspondingly, the bending efficiency increased from 0.83°/kPa to 1.28°/kPa, reflecting greater effectiveness in deformation. Although the maximum tip displacement of the previous study was higher, at 50.4 mm, compared to the 28.5 mm at 50 kPa in this work, it mainly reflects variations in finger length, and the normalized deformation behavior remains comparable.

This wider set of data enhances the reliability of performance assessments. In essence, the use of more realistic material parameters, better bending angles and efficiency, and comprehensive experimental evaluation all combine to emphasize the much better performance of the actuator developed in the present work.

5.5 Object Handling and Gripping Validation

The full gripper was tested on a variety of objects with variations in shape, surface texture, and fragility, such as eggs, paper cups, rubber balls, and tiny electronics like mouse. The gripper proved to be of high shape-conformability and was able to fit each object's shape perfectly without slipping or employing excess force. As seen on Figs. 11 and 12, the rate of grasping success was more than 95% for 20 types of objects. No damage to both the gripper and test objects was seen. The results proved that the system's ability for fragile object manipulation and safe functioning in situations involving unstructured or sensitive environments.

Author Proof

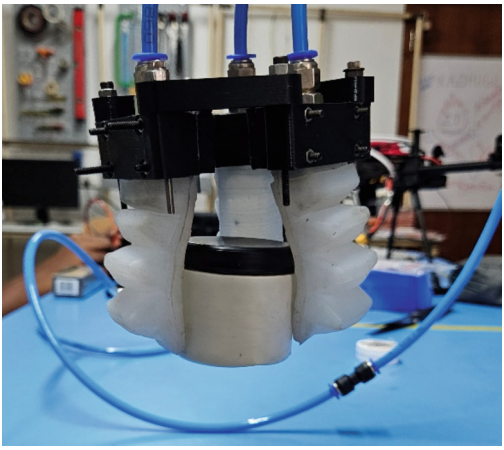


Fig. 11. Grasping Test with Grease Box (75–80 g)



Fig. 12. Grasping Test with Water Bottle (60–80 g)

6 Applications

The pneumatic soft gripper is a flexible, strong, and adaptable solution for picking objects of different shapes without damaging them and is suitable for industrial automation, packaging, food handling, electronics, farm robotics, and manipulation of delicate items. It is soft, bio-compatible materials that find application in biomedical applications such as prosthetics, rehabilitation, and surgical instruments, and its versatility finds applications in e-commerce, logistics, and collaborative robots. With modularity, strength, and quick actuation, it is superior in various applications; however, some of the challenges are

Author Proof

nonlinear pressure, bending control, absence of tactile feedback, performance limitations from air systems, fatigue at finger bases, and inaccuracies from module-based fabrication. Bio-inspired, three-claw soft pneumatic gripper is developed through a hybrid, two-stage molding technique that creates highly repeatable, airtight silicone actuators with embedded chamber geometry. Different from the conventionally purely flexible soft grippers, the proposed design achieves a unique balance between high adaptability and structural rigidity, thus allowing stable grasping of both fragile and moderately heavy objects.

7 Conclusion

This work describes the design, fabrication, modeling, and testing of a soft 3-claw pneumatic gripper, combining bio-inspired design with flexible, bio-compatible materials. The gripper was able to perform adaptive and safe grasping of multiple objects, especially fragile and irregular objects. Finite element analysis confirmed the deformation behavior under pneumatic actuation with experimental results. The modularity of the system, high repeatability, and robotic platform compatibility enable it to be ideal for industrial, medical, and collaborative robots. Future directions of the soft pneumatic gripper development may emphasize improving intelligence and responsiveness through embedded sensors like pressure, tactile, or force sensors for real-time feedback and closed-loop control. Integration with computer vision systems and investigation of high-speed actuation techniques such as electro-pneumatic systems can potentially expand its capabilities further.

AQ6

References

1. Tawk, C., Mutlu, R., Alici, G.: A 3D printed modular soft gripper integrated with metamaterials for conformal grasping. Front. Robot. AI. **8**, 799230 (2021). https://doi.org/10.3389/frobt.2021.799230
2. Georgopoulou, A., Vanderborght, B., Clemens, F.: Fabrication of a soft robotic gripper with integrated strain sensing elements using multi-material additive manufacturing. Front. Robot. AI. (2021). https://doi.org/10.3389/frobt.2021.615991
3. Fraś, J., Maciaś, M., Czubaczyński, F., Sałek, P., Główka, J.: Soft flexible gripper design, characterization and application. In: Advances in Intelligent Systems and Computing, vol. 9999, pp. 368–377. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-48923-0_40
4. Bernat, J., Gajewski, P., Kapela, R., Marcinkowska, A., Superczyńska, P.: Design, fabrication and analysis of magnetorheological soft gripper. Sensors. **22**(7), 2757 (2022). https://doi.org/10.3390/s22072757
5. Tang, Z., Lu, J., Wang, Z., Ma, G.: The development of a new variable stiffness soft gripper. Int. J. Adv. Robot. Syst. (2019). https://doi.org/10.1177/1729881419879824
6. Zhou, Y., et al.: Bio-inspired soft grippers based on impactive gripping. Adv. Sci. (2021). https://doi.org/10.1002/advs.202002017
7. Bryantono, H.D., Tsai, M.-H., Tseng, S.-C.: Design and application of soft robot grippers using low-viscosity silicone by lost-core injection molding. J. Polym. Eng. (2025). https://doi.org/10.1515/polyeng-2024-0223

Author Proof

8. Taherkhani, A., et al.: Thermally powered soft gripper covered with silver-coated nylon fabric heater reinforced with stainless steel yarn. Adv. Eng. Mater. (2024). https://doi.org/10.1002/adem.202302216
9. Ouyang, F. et al.: An optimization design method of rigid–flexible soft fingers based on dielectric elastomer actuators. Micromachines. **13**(11), 2030 (2022). https://doi.org/10.3390/mi13112030
10. Tran, N.-D., Ly, H.-H., Nguyen, X.-T., Mac, T.-T., Nguyen, A., Ta, T.D.: Hybrid gripper with passive pneumatic soft joints for grasping deformable thin objects. In: 2025 IEEE International Conference on Robotics and Automation (ICRA), pp. 7858–7864. IEEE (2025). https://doi.org/10.1109/icra55743.2025.11127915
11. Aydin, M., Sariyildiz, E., Tawk, C.D., Mutlu, R., Alici, G.: Variable stiffness improves safety and performance in soft robotics. In: 2023 IEEE International Conference on Mechatronics (ICM), pp. 1–6. IEEE (2023). https://doi.org/10.1109/icm54990.2023.10102076
12. Lei, J., Ge, Z., Fan, P., Zou, W., Jiang, T., Dong, L.: Design and manufacture of a flexible pneumatic soft gripper. Appl. Sci. **12**(13), 6306 (2022). https://doi.org/10.20944/preprints202206.0171.v1
13. Gariya, N., Asrani, A., Kumar, S., Nautiyal, H.: Bending mechanics of the soft pneumatic single air chamber bending actuator. (2023). https://doi.org/10.20944/preprints202206.0171.v1
14. Wu, Z., Li, X., Guo, Z.: A novel pneumatic soft gripper with a jointed endoskeleton structure. Chin. J. Mech. Eng. **32** (2019). https://doi.org/10.1186/s10033-019-0392-0
15. Arumugam, M., Ramalingarishi, N.D., Prabukarthi, A., Suresh, M.: Design and development of soft robotic gripper for food packaging. In: Proc. 2nd Int. Conf. Applied Mathematics and Computational Sciences (ICAMCS 2022) (2024). https://doi.org/10.1063/5.0194661

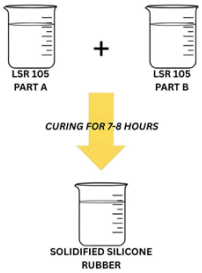
Author Proof

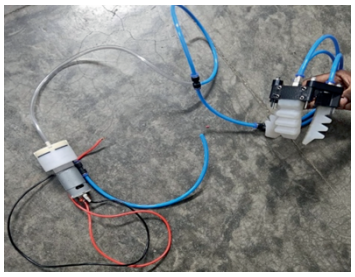
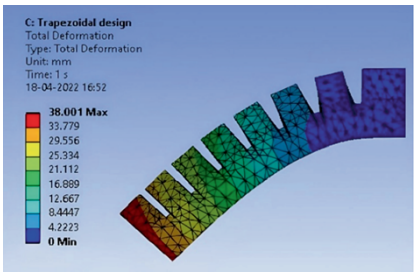
Author Queries

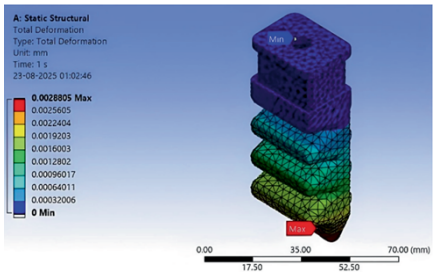
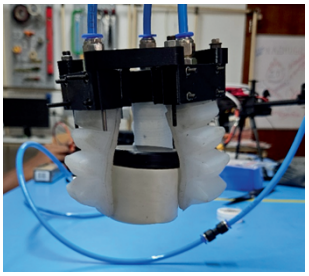
Chapter 5

Query Refs.	Details Required	Author's response
AQ1	Please check and confirm if the authors and their respective email address are correctly identified. Amend if necessary.	
AQ2	Please confirm if the corresponding author is correctly identified. Amend if necessary.	
AQ3	Please be aware that your name and affiliation and if applicable those of your co-author(s) will be published as presented in this proof. If you want to make any changes, please correct the details now. Please note that after publication corrections won't be possible. Due to data protection we standardly publish professional email addresses, but not private ones, even if they have been provided in the manuscript and are visible in this proof. If you or your co-author(s) have a different preference regarding the publication of your mail address(s) please indicate this clearly. If no changes are required for your name(s) and affiliation(s) and if you agree with the handling of email addresses, please respond with "Ok".	
AQ4	Please check and confirm if the inserted citation of Tables 1–5 is correct. If not, please suggest an alternate citation.	
AQ5	Please check and confirm if the inserted citations of Figs. 9 and 10 are correct. If not, please suggest an alternate citation.	
AQ6	Reference "15" was not cited anywhere in the text. Please provide in text citation or delete the reference from the reference list.	

Alternative Texts for Your Images, Please Check and Correct them if Required

Page no	Fig/Photo	Thumbnail	Alt-text Description
5	Fig1		A rectangular black plastic or 3D-printed component divided into three vertical sections by two internal walls, each section containing four horizontal slots or compartments stacked vertically. The component has four mounting holes, one at each corner.
6	Fig2		A rectangular object with three vertical columns of stepped, protruding shapes, each topped by a square block. The columns have a repeating pattern of horizontal ridges that decrease in width toward the bottom. The object appears to be a single molded or 3D-printed piece.
6	Fig3		A rectangular object divided into three vertical compartments by two internal walls, with a small notch centered along the top edge.
7	Fig4		The image illustrates a process where equal parts of two liquids labeled "LSR 105 Part A" and "LSR 105 Part B" are combined and then cured for seven to eight hours, resulting in solidified silicone rubber.
8	Fig5		This is a close-up of a rectangular object with three vertical compartments, each containing a series of horizontal ridges or steps. The compartments appear to be part of a mold or holder, with the rightmost compartment showing some irregularities or damage on the ridges. Two hands are holding the object on

Page no	Fig/Photo	Thumbnail	Alt-text Description
			the left and right edges.
8	Fig6		Two plastic trays with multiple compartments are placed on a newspaper. The tray on the left is orange with three long, narrow compartments. The tray on the right is black with twelve smaller, square compartments arranged in three rows of four.
9	Fig7		A mechanical gripper with two flexible, ribbed white silicone fingers attached to a black mounting plate with bolts and two blue-capped pneumatic connectors on top. The fingers are designed to close together to grasp objects.
9	Fig8		A pneumatic robotic gripper system connected by tubes to a small motor with red and black wires. The motor likely powers air flow through the tubes to the gripper, which has two soft, white, finger-like pads for grasping objects. A hand is holding the gripper end, indicating it is a tool for automated or remote manipulation.
12	Fig9		Total deformation in millimeters of a trapezoidal design after one second, with deformation values ranging from zero to a maximum of 38.001 millimeters. The deformation is highest at the left end of the design and decreases progressively toward the right end.

Page no	Fig/Photo	Thumbnail	Alt-text Description
12	Fig10		The image presents a structural analysis of a stacked object labeled "A," showing the distribution of total deformation measured in millimeters. The deformation values range from zero to approximately 0.0032 millimeters, with the highest deformation occurring at the bottom section of the stack. The deformation decreases progressively toward the top, indicating that the lower part experiences the greatest displacement under the applied conditions.
14	Fig11		A robotic gripper with three flexible, white, accordion-like fingers holds a cylindrical object. Each finger is attached to a black frame connected to blue pneumatic tubes that control the gripping mechanism. The gripper is positioned above a blue surface in a workshop environment.
14	Fig12		A robotic gripper with five soft, white, finger-like extensions is holding a transparent, flexible plastic cup. The gripper is attached to a mechanical arm above a concrete floor.